

Retrieval-based forecast adaptation

Executive summary of results

Evaluation basis. Screen and ablation averages weight equally the 12 held-out T3 dataset/window configurations (Electricity, Traffic, Solar, Exchange Rate; three (L, H) settings). Gains are nMSE improvements over the vanilla forecast; positive is better.

Decision snapshot

Question	Quantitative answer	Decision
Best baseline?	Shared full ridge, instance-space, reaches +2.733% at $K = 3$ in the screen and +2.991% at $K = 10$ in the K sweep.	Retain the vanilla-anchored ridge family.
Best gate?	Best screen gate: +0.019% ; best gate ablation: +0.045% ; diagnostic oracle: up to +13.34% .	Current learned gates do not capture the available oracle headroom.
Convex or Δ ?	Mean over six matched designs: ridge +2.427% , Δ -ridge +1.251% , convex +0.783% .	Both constraints remain positive but are dominated by ordinary ridge.
Does TS-IFA work?	The 10K ridge run reaches only +0.04% on T3; 30K ridge fits fall to -6.71% and -17.97% .	No robust success; select on the fixed T2 criterion and stop earlier.

Best average baselines and gates

Method	Retrieval	Evidence set	Mean T3 gain
Shared full ridge correction	instance, $K = 10$	K sweep	+2.991%
Shared context–future ridge correction	instance, $K = 20$	K sweep	+2.954%
Shared future–forecast ridge correction	instance, $K = 20$	K sweep	+2.861%
Shared full ridge correction	instance, $K = 3$	primary screen	+2.733%
Per-horizon Bayes context gate	instance, $K = 1$	gate ablation	+0.045%
Shared CatBoost neighbour-average gate	instance, $K = 3$	primary screen	+0.019%
Shared CatBoost context gate	instance, $K = 1$	primary screen	+0.015%
Shared neighbour-average oracle	instance, $K = 3$	diagnostic oracle	+13.34%

Winning formulas, written out

Let V_h be the vanilla forecast, C_h the context forecast, $Y_{k,h}$ the k th neighbour’s observed future, $N_{k,h}$ its backbone forecast, and $\bar{Y}_h = \sum_k w_k Y_{k,h}$. The shared coefficients below are fitted by ridge to the correction target $Y_h - V_h$ and are common to every horizon:

$$\begin{aligned}\hat{Y}_h^{\text{full}} &= V_h + \beta_V V_h + \beta_C C_h + \sum_{k=1}^K (\beta_k^Y Y_{k,h} + \beta_k^N N_{k,h}), \\ \hat{Y}_h^{\text{context+future}} &= V_h + \beta_V V_h + \beta_C C_h + \sum_{k=1}^K \beta_k^Y Y_{k,h}, \\ \hat{Y}_h^{\text{future+forecast}} &= V_h + \beta_V V_h + \sum_{k=1}^K (\beta_k^Y Y_{k,h} + \beta_k^N N_{k,h}).\end{aligned}$$

For the best gate, $\hat{Y}_h = C_h$ exactly when the T1+T2 mean of $(Y_h - V_h)^2 - (Y_h - C_h)^2$ is positive, otherwise $\hat{Y}_h = V_h$. The shared CatBoost gates use $\hat{Y}_h = A_h$ when $g_\theta(z) > 0$, otherwise V_h , where $A \in \{\bar{Y}, C\}$ and g_θ regresses the mean squared-error reduction from V to A .

Formulation ablations

Matched feature design	Ridge	Δ -ridge	Convex
Full ($C, Y_{1:K}, N_{1:K}$)	2.733%	1.515%	0.635%
$Y_{1:K}$	2.293%	1.347%	1.209%
$C + Y_{1:K}$	2.546%	1.468%	1.115%
$Y_{1:K} + N_{1:K}$	2.539%	1.339%	0.330%
$C + \bar{Y}$	2.306%	1.183%	1.063%
C	2.142%	0.652%	0.347%
Mean over designs	2.427%	1.251%	0.783%
Mean loss versus matched ridge	–	–1.176 pp	–1.643 pp

Effect of the neighbour count K

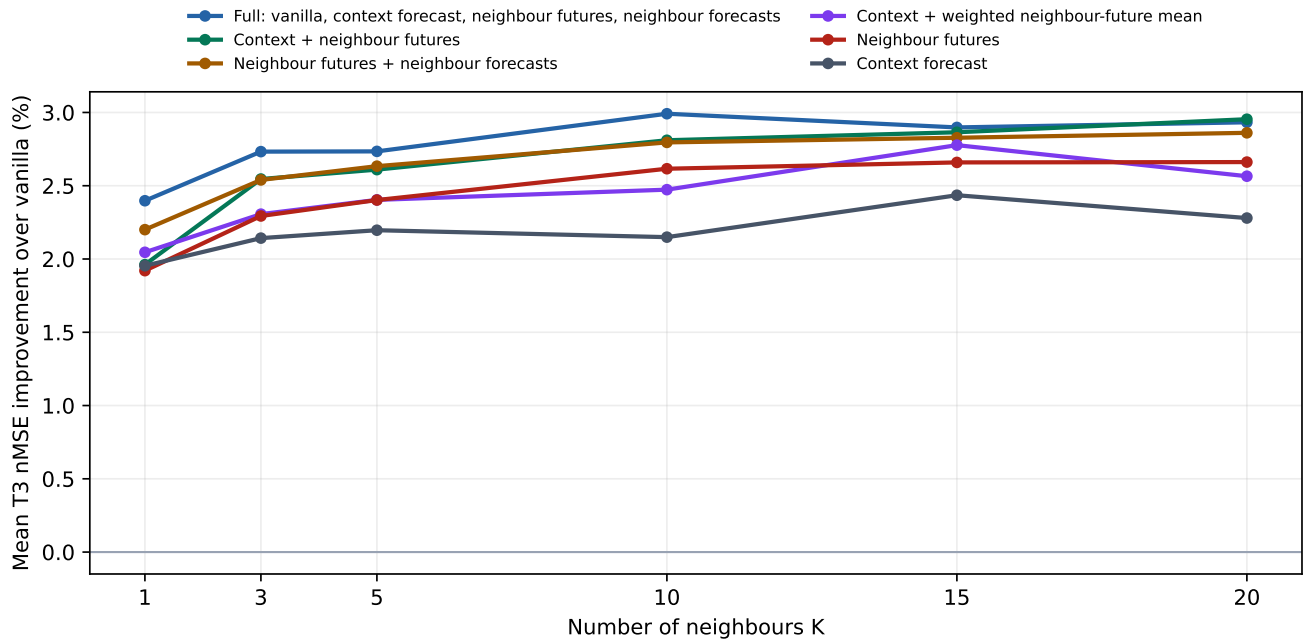


Figure 1: Mean held-out T3 nMSE improvement over 12 configurations. Only ridge baselines are shown: most improve from $K = 1$ toward $K = 10$ – 20 , with no common optimum across feature designs.

Ridge	Best K	Gain	Ridge	Best K / gain
Full	10	2.991%	$C + Y$	20 / 2.954%
Residual	20	2.861%	$C + \bar{Y}$	15 / 2.777%
Y	20	2.661%	C	15 / 2.435%

TS-IFA results

Run	T2 gain	T3 gain	Checkpoint / held-out behaviour
Joint ridge, raw, 10K	+1.84%	+0.04%	Window gains +3.26, +0.58, -3.72%; all three select 10K.
Joint ridge, raw, 30K	+3.03%	-6.71%	Selected at 29/20/20K for (168/24, 336/48, 504/168).
Joint ridge, instance, 30K	+3.25%	-17.97%	Selected at 30/26/20K for the same windows.
Joint neural, raw / instance, 30K	-	-4.60 / -2.14%	Zero T3 wins.

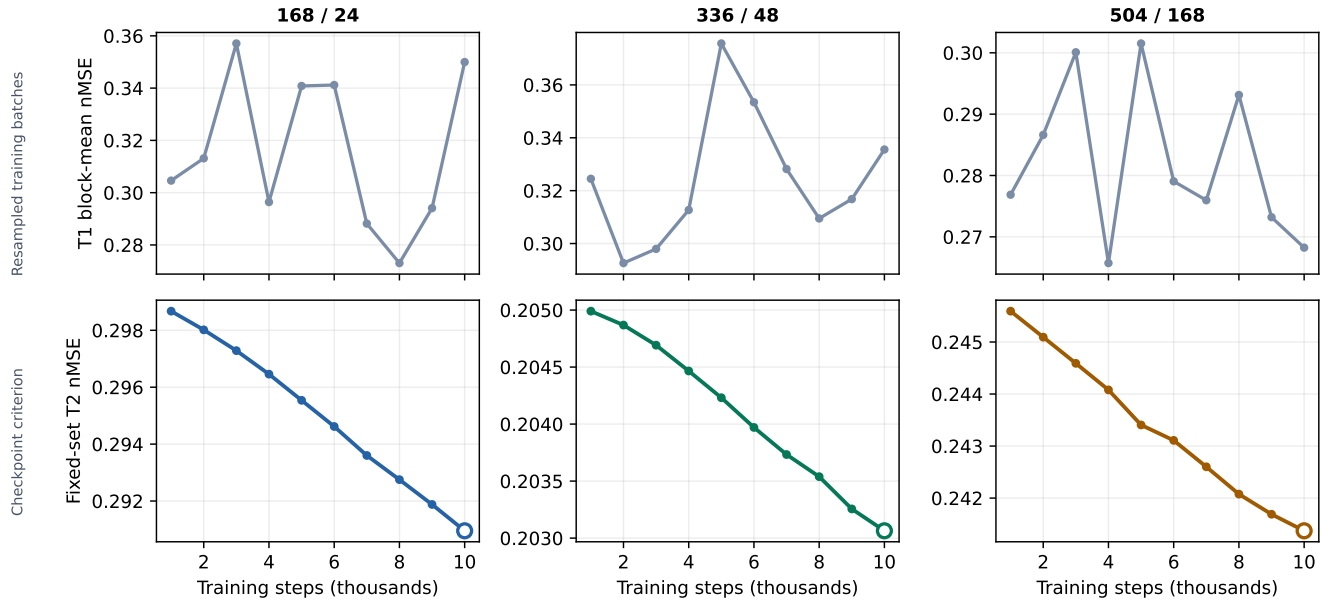


Figure 2: The 10K joint-ridge criterion curves. The gray T1 trace is a 1,000-step block mean over freshly resampled training batches, not a repeated evaluation on one fixed set, so it need not be monotone. The colored T2 trace uses a fixed set and selects 10K for all three windows.

Answer. TS-IFA does not yet work reliably. The 30K runs generalize worse than the 10K run. Four of six ridge fits select their best T2 checkpoint at 20–26K, while the two 168/24 fits select 29–30K; better T2 selection still does not translate to T3.

Decisions before reruns

Retain	Reason supported above
Ridge candidates	Reproduce the $K = 3$ screen and test $K = 10, 15, 20$ with the same held-out evaluation.
Gate diagnostics	Retain hard CatBoost/Bayes gates and oracles to measure the unclosed 13.34% diagnostic gap.
TS-IFA	Re-run with explicit T2 early stopping and untouched T3; include the 10K checkpoint and continue only while the fixed-set T2 criterion improves.